A conceptual framework for biomedical experiments

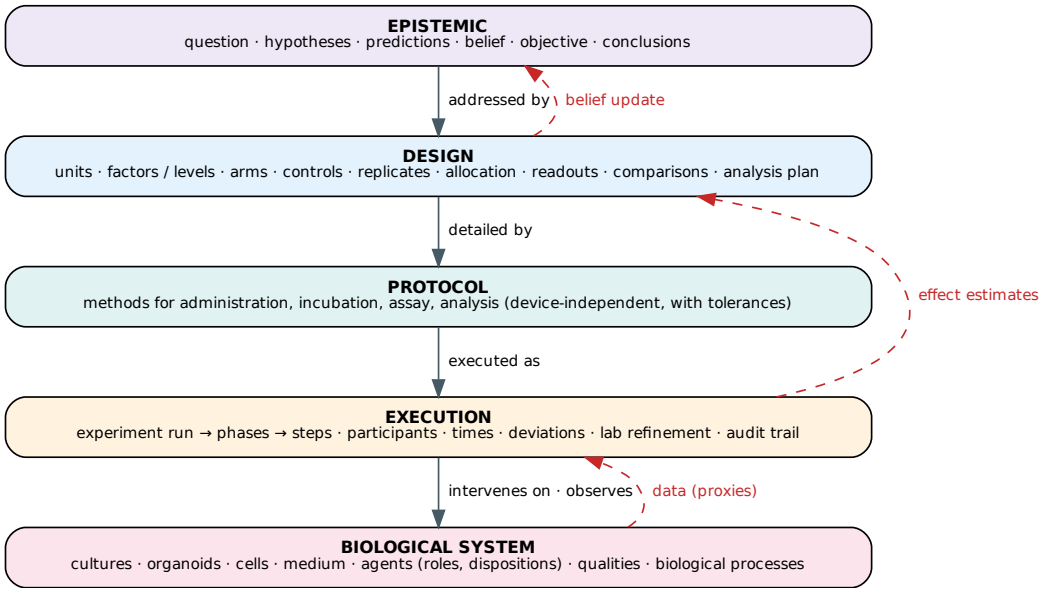
and its OWL 2 DL axiomatization, classified by a reasoner

Generated by `rsa/experiment_ontology.py`; all values ILLUSTRATIVE (D016)

I A conceptual framework for biomedical experiments

Core idea

An experiment is a planned process that intervenes on and observes a biological system, executing a design, in order to change a belief about hypotheses. The framework keeps apart what is *planned* (design, protocol), what *happens* (execution), what is *measured* (data) and what is *believed* (hypotheses, belief, conclusions); the biology is the hidden state in between, reached only through interventions and through readouts that are proxies for it. No biological fact or parameter is asserted: hypotheses *posit* biology, data are *proxies* (D016).



Core concepts

Experimental unit	what is treated independently and measured (a culture in a well)
Factor, level	what is varied (agent, dose, time of rescue) and the values it takes
Treatment	one combination of levels, given as a schedule of administrations
Arm	a treatment and the units assigned to it (membership, not parthood)
Control	an arm defined by what it controls for: negative (untreated, vehicle), positive (expected effect from prior evidence), agent-alone (the agent's own effect), comparator (isolates one factor)
Replicate	units under the same treatment; biological (independent units) vs technical (repeated measurements)
Allocation	assignment and placement of units: randomisation, blocking, avoiding edge wells, blinding
Readout	the measured quantity, its time and assay, and what it is a proxy for (size for viability)
Comparison	a pair of arms whose difference estimates the effect of exactly one factor
Interaction contrast	a difference of comparisons: does a factor's effect depend on another factor
Validity check	a comparison that licenses another (injury happened; the agent works)
Analysis plan	the decision rule of every comparison and the handling of deviations, fixed before the run
Hypothesis, prediction	a model that posits processes, and its expected effect for a comparison
Belief, information gain	a probability over hypotheses, and how much a design is expected to change it
Deviation	a difference between execution and plan; it can invalidate comparisons
Conclusion (insight)	a statement derived from effect estimates and the posterior, with provenance

Taxonomy along independent axes

System (in silico, in vitro, ex vivo, in vivo, clinical); purpose (exploratory, confirmatory, hypothesis-discriminating, dose–response, time-course); design structure (comparative, controlled, factorial, within- or between-unit, randomised, blinded); intervention (challenge, rescue or therapy, both); adequacy (validated, agent-controlled, positive-controlled, replicated, pre-specified). Classes are *defined*, so a reasoner places an experiment under every class it satisfies (Part II).

Criteria and how each is enforced

validity	criteria	enforced by
internal	single factor per comparison; negative, agent-alone and positive controls; randomisation; blinding	defined classes (reasoner); closed-world checks
statistical	biological replicates; expected information or power	closed-world counts; information-gain model
construct	readout states its proxy relation; manipulation checked (validity comparison)	ontology; licensing order in the plan
executional	identity, contamination, environment kept; plan realized; deviations recorded	run-time monitors (CRAM); audit trail
external	relevance of the system to the target biology	stated as an assumption, never inferred

Description logic for what experiments are; closed-world queries for what a design must contain; run-time monitors for what an execution must keep; probabilistic models for what it will tell us.

Lifecycle

Question → hypotheses and predictions → candidate designs → selection (information vs cost) → **adequacy check before the run** → protocol → refinement into lab actions → monitored execution → data → analysis by the pre-specified rules → belief update → conclusions → next question. Foundations: BFO-style upper level, OBI patterns (planned process, study design, assay), Cell 01's biology ontology for the domain.

II The axiomatization

What Part II is

The challenge-rescue experiment axiomatized as a biomedical experiment in $\mathcal{SROIQ}(\mathbf{D})$ (OWL 2 DL). One source gives the DL notation below, an OWL ontology (`experiment.ofn`, `experiment.owl`) and the input of the HermiT reasoner, which checks consistency, classifies the concepts and realizes the individuals. Compared with the first axiomatization, the design is a *plan* that the experiment executes; cultures are *units assigned to arms* (membership, not parthood); agents are chemical entities bearing *roles* and *dispositions*; arms, comparisons, designs and experiments are *defined* concepts that the reasoner classifies from their content; the biomedical adequacy of a design is part of the hierarchy; hypotheses are information entities that *posit* processes, so nothing biological is asserted as fact; size is a *proxy* for viability. Concepts are set in sans, roles in *italics*, individuals in `mono`.

1 Upper level

A BFO-style upper level. Continuants (material entities, qualities, realizable entities, information content entities) and occurrents (processes) are disjoint; planned processes are distinct from biological processes.

MaterialEntity ⊑ Continuant

SpecificallyDependentContinuant ⊑ Continuant

Quality ⊑ SpecificallyDependentContinuant

RealizableEntity ⊑ SpecificallyDependentContinuant

InformationContentEntity ⊑ Continuant

Process ⊑ Occurrent

Continuant ⊓ Occurrent ⊑ ⊥

Disjoint(MaterialEntity, SpecificallyDependentContinuant, InformationContentEntity)

Quality ⊓ RealizableEntity ⊑ ⊥

Role ⊑ RealizableEntity

Disposition ⊑ RealizableEntity

Role ⊓ Disposition ⊑ ⊥

PlannedProcess ⊑ Process

BiologicalProcess ⊑ Process

PlannedProcess ⊓ BiologicalProcess ⊑ ⊥

PlanSpecification ⊑ InformationContentEntity

DataItem ⊑ InformationContentEntity

Hypothesis ⊑ InformationContentEntity

Objective ⊑ InformationContentEntity

Conclusion ⊑ InformationContentEntity

Disjoint(PlanSpecification, DataItem, Hypothesis, Objective, Conclusion)

2 Relations and partonomy

Parthood is transitive and kind-preserving: processes have processes as parts, material entities material entities, information entities information entities. Its kinds are simple sub-roles (so they may be counted). Location, membership, participation, execution and bearing are not parthood.

Trans(*hasPart*)

$partOf \equiv hasPart^{-}$

$hasPhase \sqsubseteq hasPart$

$hasStep \sqsubseteq hasPart$

$hasWell \sqsubseteq hasPart$

$hasOrganoid \sqsubseteq hasPart$

$hasCell \sqsubseteq hasPart$

$hasMedium \sqsubseteq hasPart$

$hasArm \sqsubseteq hasPart$

$hasComparisonSpec \sqsubseteq hasPart$

$hasAdministrationSpec \sqsubseteq hasPart$

$hasObjective \sqsubseteq hasPart$

Process ⊑ ∀*hasPart*.Process

MaterialEntity ⊑ ∀*hasPart*.MaterialEntity

InformationContentEntity ⊑ ∀*hasPart*.InformationContentEntity

$containedIn \equiv contains^{-}$

$containedIn \circ partOf \sqsubseteq locatedIn$

$participatesIn \equiv hasParticipant^{-}$

$hasPart \circ hasParticipant \sqsubseteq hasParticipant$

$hasSpecifiedInput \sqsubseteq hasParticipant$

$inheresIn \equiv bearerOf^{-}$

Func($inheresIn$)

$hasAssignedUnit \equiv assignedTo^{-}$

Func($assignedTo$)

Func($usesAgent$)

Trans($precedes$)

$\exists executes.\top \sqsubseteq$ PlannedProcess

$\top \sqsubseteq \forall executes.$PlanSpecification

$\exists isAbout.\top \sqsubseteq$ InformationContentEntity

$\exists bearerOf.\top \sqsubseteq$ Continuant

$\top \sqsubseteq \forall bearerOf.$SpecificallyDependentContinuant

$\exists realizedIn.\top \sqsubseteq$ RealizableEntity

$\top \sqsubseteq \forall realizedIn.$Process

Func($dose_a u$)

Func($atTime_h$)

Func($fromTime_h$)

Func($toTime_h$)

Func($expectedInformationGain_b its$)

Func($radius_u m$)

Func($wellRow$)

Func($wellColumn$)

3 Biological material and labware

An organoid culture is organoids in medium; an organoid has cells. A well holds at most one culture; a culture is in a well, not part of it. Chemical agents bear roles in the experiment (challenge, rescue, vehicle) and dispositions (toxicity); size is a quality of organoids, viability and recovery capacity are dispositions of cultures.

OrganoidCulture $\sqsubseteq$ MaterialEntity $\sqcap \exists hasOrganoid.$Organoid $\sqcap \exists hasMedium.$CultureMedium

Organoid $\sqsubseteq$ MaterialEntity $\sqcap \exists hasCell.$Cell

Cell $\sqsubseteq$ MaterialEntity

CultureMedium $\sqsubseteq$ MaterialEntity

ChemicalAgent $\sqsubseteq$ MaterialEntity

Plate $\sqsubseteq$ MaterialEntity $\sqcap \exists hasWell.$Well

Well $\sqsubseteq$ MaterialEntity $\sqcap (\leq 1 contains.$OrganoidCulture)

Disjoint(OrganoidCulture, Organoid, Cell, CultureMedium, ChemicalAgent, Plate, Well)

OrganoidSize $\sqsubseteq$ Quality $\sqcap \exists inheresIn.$Organoid

Viability $\sqsubseteq$ Disposition $\sqcap \exists inheresIn.$OrganoidCulture $\sqcap \forall realizedIn.$CultureGrowth

RecoveryCapacity $\sqsubseteq$ Disposition $\sqcap \exists inheresIn.$OrganoidCulture $\sqcap \forall realizedIn.$RecoveryProcess

ToxicityDisposition $\sqsubseteq$ Disposition $\sqcap \exists inheresIn.$ChemicalAgent $\sqcap \forall realizedIn.$InjuryProcess

Toxicant $\equiv$ ChemicalAgent $\sqcap \exists bearerOf.$ToxicityDisposition

ChallengeAgentRole $\sqsubseteq$ Role

RescueAgentRole $\sqsubseteq$ Role

VehicleRole $\sqsubseteq$ Role

ExperimentalUnitRole $\sqsubseteq$ Role

Disjoint(ChallengeAgentRole, RescueAgentRole, VehicleRole, ExperimentalUnitRole)

ChallengeAgent $\equiv$ ChemicalAgent $\sqcap \exists bearerOf.$ChallengeAgentRole

RescueAgent $\equiv$ ChemicalAgent $\sqcap \exists bearerOf.$RescueAgentRole

Vehicle $\equiv$ ChemicalAgent $\sqcap \exists bearerOf.$VehicleRole

Disjoint(ChallengeAgent, RescueAgent, Vehicle)

ChallengeAgent $\sqsubseteq$ Toxicant

the challenge role is given to an agent because of its toxicity

ExperimentalUnit $\equiv$ OrganoidCulture $\sqcap \exists bearerOf.$ExperimentalUnitRole

4 Biological processes

Processes in which the cultures participate. A loss of recovery capacity (a point of no return) is a process type that hypotheses may posit; nothing here asserts that it occurs.

InjuryProcess $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

ProtectiveResponse $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

RecoveryProcess $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

CellDeath $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

CultureGrowth $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

LossOfRecoveryCapacity $\sqsubseteq$ BiologicalProcess $\sqcap \exists hasParticipant$.OrganoidCulture

Disjoint(InjuryProcess, ProtectiveResponse, RecoveryProcess, CellDeath, CultureGrowth)

5 Plans: designs, arms, administrations, comparisons

A study design has arms and comparisons; an arm specifies the administrations its units receive (agent, dose, time). Arm types are defined by what is administered, comparison types by the arms compared, design types by their arms and comparisons; the reasoner classifies them. Replication and the information target are ILLUSTRATIVE design criteria.

StudyDesign $\sqsubseteq$ PlanSpecification $\sqcap \exists hasObjective$.Objective $\sqcap \exists hasArm$.ArmSpecification

ArmSpecification $\sqsubseteq$ PlanSpecification $\sqcap \forall hasAdministrationSpec$.AdministrationSpecification

AdministrationSpecification $\sqsubseteq$ PlanSpecification $\sqcap (= 1 usesAgent$.ChemicalAgent$) \sqcap \exists dose_a u$.decimal $\sqcap$ $\exists atTime_h$.decimal

ComparisonSpecification $\sqsubseteq$ PlanSpecification $\sqcap (= 2 compares$.ArmSpecification$)$

Disjoint(StudyDesign, ArmSpecification, AdministrationSpecification, ComparisonSpecification)

ChallengeSpec $\equiv$ AdministrationSpecification $\sqcap \exists usesAgent$.ChallengeAgent

RescueSpec $\equiv$ AdministrationSpecification $\sqcap \exists usesAgent$.RescueAgent

VehicleSpec $\equiv$ AdministrationSpecification $\sqcap \exists usesAgent$.Vehicle

UntreatedArm $\equiv$ ArmSpecification $\sqcap \neg \exists hasAdministrationSpec$.AdministrationSpecification

VehicleControlArm $\equiv$ ArmSpecification $\sqcap \exists hasAdministrationSpec$.VehicleSpec $\sqcap$ $\neg \exists hasAdministrationSpec$.(ChallengeSpec $\sqcup$ RescueSpec)

NegativeControlArm $\equiv$ UntreatedArm $\sqcup$ VehicleControlArm

ChallengeOnlyArm $\equiv$ ArmSpecification $\sqcap \exists hasAdministrationSpec$.ChallengeSpec $\sqcap$ $\neg \exists hasAdministrationSpec$.RescueSpec

RescueOnlyArm $\equiv$ ArmSpecification $\sqcap \exists hasAdministrationSpec$.RescueSpec $\sqcap \neg \exists hasAdministrationSpec$.ChallengeSpec

ChallengeRescueArm $\equiv$ ArmSpecification $\sqcap \exists hasAdministrationSpec$.ChallengeSpec $\sqcap$ $\exists hasAdministrationSpec$.RescueSpec

ReplicatedArm $\equiv$ ArmSpecification $\sqcap (\geq 3 hasAssignedUnit$.ExperimentalUnit$)$

InjuryValidityComparison $\equiv$ ComparisonSpecification $\sqcap \exists compares$.ChallengeOnlyArm $\sqcap \exists compares$.NegativeControlArm

RescueEffectComparison $\equiv$ ComparisonSpecification $\sqcap \exists compares$.ChallengeRescueArm $\sqcap$ $\exists compares$.ChallengeOnlyArm

AgentAloneComparison $\equiv$ ComparisonSpecification $\sqcap \exists compares$.RescueOnlyArm $\sqcap \exists compares$.NegativeControlArm

RescueTimingComparison $\equiv$ ComparisonSpecification $\sqcap (= 2 compares$.ChallengeRescueArm$)$

DiscriminationObjective $\equiv$ Objective $\sqcap (\geq 2 isAbout$.Hypothesis$)$

6 Factors, levels and treatments

A factor is what the experimenter varies (here: the challenge, the time of the rescue); its levels are the values it takes, each specifying administrations. An arm's treatment is one level of every factor (atLevel; a level is shared by arms, so this is not parthood). A comparison contrasts the factors in which its two arms differ; a well-formed comparison contrasts exactly one. Which factors a comparison contrasts, and whether an arm's schedule is the union of its levels, are computed under the closed world and asserted.

Factor $\sqsubseteq$ PlanSpecification $\sqcap$ ($\geq 2 \mathit{hasLevel}$.FactorLevel)

FactorLevel $\sqsubseteq$ PlanSpecification $\sqcap$ ($= 1 \mathit{levelOf}$.Factor)

$\mathit{levelOf} \equiv \mathit{hasLevel}^{-}$

$\mathit{hasLevel} \sqsubseteq \mathit{hasPart}$

$\mathit{hasFactor} \sqsubseteq \mathit{hasPart}$

ArmSpecification $\sqsubseteq \forall \mathit{atLevel}$.FactorLevel

ComparisonSpecification $\sqsubseteq \forall \mathit{contrasts}$.Factor

SingleFactorComparison $\equiv$ ComparisonSpecification $\sqcap$ ($= 1 \mathit{contrasts}$.Factor)

FactorialDesign $\equiv$ StudyDesign $\sqcap$ ($\geq 2 \mathit{hasFactor}$.Factor)

WellContrastedDesign $\equiv$ StudyDesign $\sqcap \forall \mathit{hasComparisonSpec}$.SingleFactorComparison

InteractionContrast $\sqsubseteq$ PlanSpecification $\sqcap$ ($= 2 \mathit{differenceOf}$.ComparisonSpecification)

$\mathit{hasInteractionContrast} \sqsubseteq \mathit{hasPart}$

InteractionAwareDesign $\equiv$ StudyDesign $\sqcap \exists \mathit{hasInteractionContrast}$.InteractionContrast

Disjoint(Factor, FactorLevel, StudyDesign, ArmSpecification, AdministrationSpecification, ComparisonSpecification, InteractionContrast)

7 Controls

A control is defined by what it controls for. Negative controls (untreated, vehicle) give the baseline; the agent-alone arm separates the rescue agent's own effect from rescue of injury; a positive control is an arm whose outcome is expected from prior evidence, so that a null result elsewhere can be interpreted (the agent and the assay worked in this run).

Expectation $\sqsubseteq$ DataItem

PriorEvidence $\sqsubseteq$ DataItem

EvidenceBasedExpectation $\equiv$ Expectation $\sqcap \exists \mathit{derivedFrom}$.PriorEvidence

PositiveControlArm $\equiv$ ArmSpecification $\sqcap \exists \mathit{hasExpectedOutcome}$.EvidenceBasedExpectation

PositiveControlComparison $\equiv$ ComparisonSpecification $\sqcap \exists \mathit{compares}$.PositiveControlArm $\sqcap$ $\exists \mathit{compares}$.ChallengeOnlyArm

PositiveControlledDesign $\equiv$ StudyDesign $\sqcap \exists \mathit{hasComparisonSpec}$.PositiveControlComparison

8 Readouts and analysis plan

A readout (endpoint) specifies what is measured, when and by which assay, and what it is a proxy for. The analysis plan, part of the design and fixed before the run, has one decision rule per comparison (a threshold test, a Bayesian update over the hypotheses, or an estimate) and a rule for deviations. A design is pre-specified when every comparison is covered by a rule.

$\mathit{hasReadout} \sqsubseteq \mathit{hasPart}$

$\mathit{hasAnalysisPlan} \sqsubseteq \mathit{hasPart}$

$\mathit{hasDecisionRule} \sqsubseteq \mathit{hasPart}$

ReadoutSpecification $\sqsubseteq$ PlanSpecification $\sqcap \exists \mathit{measures}$.Quality $\sqcap \exists \mathit{atTime}_h$.decimal

SizeReadout ≡ ReadoutSpecification ⊓ ∃*measures*.OrganoidSize ⊓ ∃*isProxyFor*.Viability

AnalysisPlan ⊑ PlanSpecification ⊓ ∃*hasDecisionRule*.DecisionRule

DecisionRule ⊑ PlanSpecification

ComparisonRule ⊑ DecisionRule ⊓ (= 1*appliesTo*.ComparisonSpecification)

ThresholdRule ⊑ ComparisonRule

EstimationRule ⊑ ComparisonRule

BayesianUpdateRule ⊑ ComparisonRule ⊓ (≥ 2*isAbout*.Hypothesis)

DeviationRule ⊑ DecisionRule ⊓ ¬∃*appliesTo*.⊤

Disjoint(ThresholdRule, EstimationRule, BayesianUpdateRule, DeviationRule)

Disjoint(ReadoutSpecification, AnalysisPlan, DecisionRule, Factor, StudyDesign, ArmSpecification, ComparisonSpecification)

CoveredComparison ≡ ComparisonSpecification ⊓ ∃*appliesTo*$^{-}$.ComparisonRule

PreSpecifiedDesign ≡ StudyDesign ⊓ ∃*hasReadout*.ReadoutSpecification ⊓ ∃*hasAnalysisPlan*.AnalysisPlan ⊓ ∀*hasComparisonSpec*.CoveredComparison

DeviationAwareDesign ≡ PreSpecifiedDesign ⊓ ∃*hasAnalysisPlan*.∃*hasDecisionRule*.DeviationRule

9 Design classes

Design types are defined by their arms, comparisons, factors and analysis plan; the reasoner places every design under each class it satisfies. Adequate control (the reviewer's standard) requires the agent-alone and vehicle controls, a positive control, replication, single-factor comparisons and a pre-specified analysis. Replication and the information target are ILLUSTRATIVE criteria.

ComparativeDesign ≡ StudyDesign ⊓ (≥ 2*hasArm*.ArmSpecification) ⊓ ∃*hasComparisonSpec*.ComparisonSpecification

ControlledDesign ≡ ComparativeDesign ⊓ ∃*hasArm*.NegativeControlArm

InjuryRescueDesign ≡ ControlledDesign ⊓ ∃*hasArm*.ChallengeRescueArm ⊓ ∃*hasArm*.ChallengeOnlyArm ⊓ ∃*hasComparisonSpec*.RescueEffectComparison

ValidatedInjuryRescueDesign ≡ InjuryRescueDesign ⊓ ∃*hasComparisonSpec*.InjuryValidityComparison

AgentControlledInjuryRescueDesign ≡ ValidatedInjuryRescueDesign ⊓ ∃*hasArm*.RescueOnlyArm ⊓ ∃*hasArm*.VehicleControlArm ⊓ ∃*hasComparisonSpec*.AgentAloneComparison

TherapeuticWindowDesign ≡ InjuryRescueDesign ⊓ (≥ 2*hasArm*.ChallengeRescueArm)

ReplicatedDesign ≡ StudyDesign ⊓ ∀*hasArm*.ReplicatedArm

AdequatelyControlledInjuryRescueDesign ≡ AgentControlledInjuryRescueDesign ⊓ PositiveControlledDesign ⊓ ReplicatedDesign ⊓ WellContrastedDesign ⊓ PreSpecifiedDesign

HypothesisDiscriminatingDesign ≡ StudyDesign ⊓ ∃*hasObjective*.DiscriminationObjective

SufficientlyInformativeDesign ≡ StudyDesign ⊓ ∃*expectedInformationGain*$_b$*its*.decimal$_{\geq 0.5}$

10 Experiments, phases and steps

An experiment (investigation) is a planned process that executes exactly one study design; its classification follows its design's. Its parts are three phases, their parts the steps: administrations and incubations, assays, data transformations.

Investigation ⊑ PlannedProcess ⊓ (= 1*executes*.StudyDesign)

ComparativeExperiment ≡ Investigation ⊓ ∃*executes*.ComparativeDesign

ControlledExperiment ≡ Investigation ⊓ ∃*executes*.ControlledDesign

InjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.InjuryRescueDesign

ValidatedInjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.ValidatedInjuryRescueDesign

AdequatelyControlledInjuryRescueExperiment ≡ Investigation ⊓ ∃*executes*.AdequatelyControlledInjuryRescueDesign

TherapeuticWindowExperiment ≡ Investigation ⊓ ∃*executes*.TherapeuticWindowDesign

FactorialExperiment ≡ Investigation ⊓ ∃*executes*.FactorialDesign

HypothesisDiscriminatingExperiment ≡ Investigation ⊓ ∃*executes*.HypothesisDiscriminatingDesign

ComparativeExperiment ⊑ (= 1*hasPhase*.CultivationPhase) ⊓ (= 1*hasPhase*.ObservationPhase) ⊓ (= 1*hasPhase*.AnalysisPhase)

Phase ⊑ PlannedProcess

CultivationPhase ⊑ Phase ⊓ ∀*hasStep*.(Administration ⊔ Incubation)

ObservationPhase ⊑ Phase ⊓ ∀*hasStep*.Assay

AnalysisPhase ⊑ Phase ⊓ ∀*hasStep*.DataTransformation

Disjoint(CultivationPhase, ObservationPhase, AnalysisPhase)

Administration ⊑ PlannedProcess ⊓ (= 1*executes*.AdministrationSpecification) ⊓ ∃*hasSpecifiedInput*.ChemicalAgent ⊓ ∃*hasSpecifiedInput*.ExperimentalUnit

ChallengeAdministration ≡ Administration ⊓ ∃*executes*.ChallengeSpec

RescueAdministration ≡ Administration ⊓ ∃*executes*.RescueSpec

Incubation ⊑ PlannedProcess ⊓ ∃*hasParticipant*.OrganoidCulture

Assay ⊑ PlannedProcess ⊓ ∃*hasSpecifiedInput*.ExperimentalUnit ⊓ ∃*hasSpecifiedOutput*.DataItem

ImagingAssay ⊑ Assay ⊓ ∃*hasSpecifiedOutput*.Image

DataTransformation ⊑ PlannedProcess

ImageAnalysis ⊑ DataTransformation ⊓ ∃*hasSpecifiedInput*.Image ⊓ ∃*hasSpecifiedOutput*.SizeDatum

StatisticalComparison ⊑ DataTransformation ⊓ (= 1*executes*.ComparisonSpecification) ⊓ ∃*executes*.ComparisonRule ⊓ ∃*hasSpecifiedOutput*.EffectEstimate

ModelUpdate ⊑ DataTransformation ⊓ ∃*hasSpecifiedOutput*.PosteriorDistribution

Disjoint(Phase, Administration, Incubation, Assay, DataTransformation)

11 Data, hypotheses and conclusions

Images are about units; a size datum is about organoid size and is a proxy for viability. Recovery hypotheses are about recovery processes: the graded one posits no loss of recovery capacity, the threshold one posits it, the protective one posits a protective response that delays it. Predictions derive from hypotheses; conclusions (the insights) derive from effect estimates and posteriors.

Image ⊑ DataItem ⊓ ∃*isAbout*.ExperimentalUnit

MeasurementDatum ⊑ DataItem

SizeDatum ⊑ MeasurementDatum ⊓ ∃*isAbout*.OrganoidSize ⊓ ∃*isProxyFor*.Viability

EffectEstimate ⊑ DataItem ⊓ ∃*derivedFrom*.SizeDatum

PosteriorDistribution ⊑ DataItem ⊓ (≥ 2*isAbout*.Hypothesis)

Prediction ⊑ Expectation ⊓ ∃*derivedFrom*.Hypothesis ⊓ ∃*isAbout*.ComparisonSpecification

Disjoint(Image, MeasurementDatum, EffectEstimate, PosteriorDistribution, Expectation, PriorEvidence)

RecoveryHypothesis ⊑ Hypothesis ⊓ ∃*isAbout*.RecoveryProcess

GradedRecoveryHypothesis ⊑ RecoveryHypothesis ⊓ ¬∃*posits*.LossOfRecoveryCapacity

ThresholdRecoveryHypothesis ⊑ RecoveryHypothesis ⊓ ∃*posits*.LossOfRecoveryCapacity

ProtectiveResponseHypothesis ⊑ ThresholdRecoveryHypothesis ⊓ ∃*posits*.(ProtectiveResponse ⊓ ∃*delays*.LossOfRecoveryCapacity)

Conclusion ⊑ ∃*derivedFrom*.(EffectEstimate ⊔ PosteriorDistribution)

12 Laboratories, devices and capabilities

A laboratory has devices as parts; a device bears functions, dispositions realized only in processes of one kind (a dosing function only in administrations, an imaging function only in imaging assays). A laboratory's capability classes are defined by the functions its devices bear, by the environments it provides and by its validated readout bindings, so the reasoner classifies laboratories from their devices. Reagent stocks are located in a laboratory, not parts of it.

Laboratory $\sqsubseteq$ MaterialEntity $\sqcap \exists hasDevice.$Device

$hasDevice \sqsubseteq hasPart$

Device $\sqsubseteq$ MaterialEntity

Disjoint(Laboratory, Device, OrganoidCulture, ChemicalAgent, Plate, Well)

DeviceFunction $\sqsubseteq$ Disposition $\sqcap \exists inheresIn.$Device

DosingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn.$Administration

IncubationFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn.$Incubation

BrightfieldImagingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn.$ImagingAssay

LuminescenceReadingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn.$LuminescenceAssay

AbsorbanceReadingFunction $\sqsubseteq$ DeviceFunction $\sqcap \forall realizedIn.$AbsorbanceAssay

CalibrationFunction $\sqsubseteq$ DeviceFunction

QuarantineFunction $\sqsubseteq$ DeviceFunction

TransportFunction $\sqsubseteq$ DeviceFunction

PlateHandlingFunction $\sqsubseteq$ DeviceFunction

StorageFunction $\sqsubseteq$ DeviceFunction

EndpointAssay $\sqsubseteq$ Assay

LuminescenceAssay $\sqsubseteq$ EndpointAssay

AbsorbanceAssay $\sqsubseteq$ EndpointAssay

ImagingAssay $\sqcap$ EndpointAssay $\sqsubseteq \bot$

MetabolicActivity $\sqsubseteq$ Quality $\sqcap \exists inheresIn.$OrganoidCulture

EnvironmentProfile $\sqsubseteq$ PlanSpecification

ReadoutBinding $\sqsubseteq$ InformationContentEntity $\sqcap \exists isAbout.$Quality

ValidatedReadoutBinding $\sqsubseteq$ ReadoutBinding

DosingLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$DosingFunction

CultivationLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$IncubationFunction

BrightfieldImagingLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$BrightfieldImagingFunction

LuminescenceReadoutLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$LuminescenceReadingFunction

AbsorbanceReadoutLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$AbsorbanceReadingFunction

QualityControlledLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$CalibrationFunction

ContainmentCapableLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$QuarantineFunction

ConnectedLab $\equiv$ Laboratory $\sqcap \exists hasDevice.\exists bearerOf.$TransportFunction

MultiEnvironmentLab $\equiv$ Laboratory $\sqcap (\geq 2\, providesEnvironment.$EnvironmentProfile)

ValidatedSizeReadoutLab $\equiv$ Laboratory $\sqcap \exists hasReadoutBinding.$(ValidatedReadoutBinding $\sqcap \exists isAbout.$OrganoidSize)

ValidatedMetabolicReadoutLab $\equiv$ Laboratory $\sqcap \exists hasReadoutBinding.$(ValidatedReadoutBinding $\sqcap$ $\exists isAbout.$MetabolicActivity)

TreatmentAndImagingModule $\equiv$ DosingLab $\sqcap$ CultivationLab $\sqcap$ BrightfieldImagingLab $\sqcap$ ValidatedSizeReadoutLab

13 Design requirements

What a design needs from a laboratory follows from its content: administrations need dosing, an imaging readout needs imaging, an endpoint readout needs the corresponding reader and comes after every non-destructive observation, an environment factor needs as many controlled environments as it has levels. Requirement classes are defined, so the reasoner classifies designs by their needs. Each requirement is met by one laboratory capability class; whether the laboratories of a site meet all of a design's requirements, with its reagents, readout bindings and timing, is a closed-world question over the individuals.

RequiresDosing ≡ StudyDesign ⊓ ∃*hasArm*.∃*hasAdministrationSpec*.AdministrationSpecification

RequiresCultivation ≡ StudyDesign ⊓ ∃*hasArm*.ArmSpecification

ImagingReadout ⊑ ReadoutSpecification

EndpointReadout ⊑ ReadoutSpecification

LuminescenceReadout ⊑ EndpointReadout

AbsorbanceReadout ⊑ EndpointReadout

ImagingReadout ⊓ EndpointReadout ⊑ ⊥

RequiresImaging ≡ StudyDesign ⊓ ∃*hasReadout*.ImagingReadout

RequiresLuminescence ≡ StudyDesign ⊓ ∃*hasReadout*.LuminescenceReadout

RequiresAbsorbance ≡ StudyDesign ⊓ ∃*hasReadout*.AbsorbanceReadout

RequiresEndpointHandling ≡ StudyDesign ⊓ ∃*hasReadout*.EndpointReadout

EnvironmentFactor ⊑ Factor

RequiresMultipleEnvironments ≡ StudyDesign ⊓ ∃*hasFactor*.EnvironmentFactor

14 Alignment to Cell 01's biology ontology

Where a concept means the same as one of `aicor-biology.ttl` (read only), it is subsumed by it. Joint reasoning: consistent, 0 unsatisfiable concepts, HermiT 0.63 s.

Organoid ⊑ bio : Organoid

CultureMedium ⊑ bio : CultureMedium

Toxicant ⊑ bio : Toxicant

RescueAgent ⊑ bio : RescueAgent

ChallengeAdministration ⊑ bio : ControlledToxicChallenge

RescueAdministration ⊑ bio : RescueIntervention

ImagingAssay ⊑ bio : ImagingProcess

Incubation ⊑ bio : IncubationProcess

Hypothesis ⊑ bio : Hypothesis

Prediction ⊑ bio : Prediction

ProtectiveResponse ⊑ bio : ProtectiveResponse

RecoveryProcess ⊑ bio : RecoveryProcess

CellDeath ⊑ bio : CellDeathProcess

CultureGrowth ⊑ bio : CultureGrowth

InjuryProcess ⊑ bio : DamageAccumulation

Plate ⊑ bio : CulturePlate

Well ⊑ bio : CultureWell

15 What OWL 2 DL cannot state: closed-world checks

Single difference compares two arms' administration schedules; replication counts units; edge placement and one-plate placement compare values reached by different paths; whether every specification was executed needs the closed world. These are queries over the reasoned ABox (Section 16).

16 Reasoning results

The challenge-rescue experiment of experiment.ldl, as derived and placed, with the surrogate run's results (ILLUSTRATIVE, D016)

HermiT: consistent, 0 unsatisfiable concepts, 0.57 s.

individual	inferred most specific concepts
challenge_rescue_design	FactorialDesign, HypothesisDiscriminatingDesign, PreSpecifiedDesign, RequiresImaging, SufficientlyInformativeDesign, ValidatedInjuryRescueDesign, WellContrastedDesign
objective	DiscriminationObjective
agent_challenge	ChallengeAgent
agent_rescue	RescueAgent
arm_rescued	ChallengeRescueArm
arm_comparator	ChallengeOnlyArm
arm_control	UntreatedArm
cmp_validity	CoveredComparison, InjuryValidityComparison, SingleFactorComparison
cmp_primary	CoveredComparison, RescueEffectComparison, SingleFactorComparison
experiment_challenge_rescue	FactorialExperiment, HypothesisDiscriminatingExperiment, ValidatedInjuryRescueExperiment
step_x1	ChallengeAdministration
step_x3	RescueAdministration
step_a1	ModelUpdate, StatisticalComparison

Closed-world checks:

- pass: every arm has one level of every factor (3 arms × 2 factors)
- pass: single factor: validity (comparator vs control differ in challenge)
- pass: schedules matched: validity (administrations differ at 0 h, all belonging to the contrasted factor)
- pass: single factor: primary (rescued vs comparator differ in rescue)
- pass: schedules matched: primary (administrations differ at 48 h, all belonging to the contrasted factor)
- information: full factorial (information, not required) (3 of 4 level combinations; missing: challenge=none, rescue=0p7au_48h)
- fail: interaction estimable (no complete 2×2 sub-table)
- pass: every comparison has a decision rule (2 of 2)
- fail: positive control licenses the primary comparison (no positive-control arm)
- fail: replicates ≥ 3 per arm (rescued 1, comparator 1, control 1)
- fail: units off the plate edge (edge wells: ['A01', 'A02', 'A05'])
- pass: all units on one plate (plate_P17)
- pass: each administration specification executed once (3 administrations for 3 specifications)

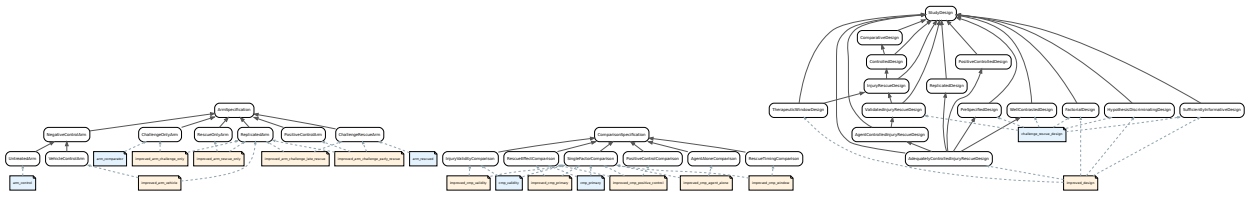
ILLUSTRATIVE improved design (not run): a challenge x time-of-rescue factorial, vehicle-matched, rescue-only and early-rescue (positive control) arms, 3 replicates per arm, randomised interior wells, pre-specified analysis; information gain for n = 3 from the same surrogate (primary contrast only)

HermiT: consistent, 0 unsatisfiable concepts, 0.74 s.

individual	inferred most specific concepts
`improved_design`	AdequatelyControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, RequiresImaging, SufficientlyInformativeDesign, TherapeuticWindowDesign
`improved_objective`	DiscriminationObjective
`agent_challenge`	ChallengeAgent
`agent_rescue`	RescueAgent
`agent_vehicle`	Vehicle
`improved_arm_vehicle`	ReplicatedArm, VehicleControlArm
`improved_arm_rescue_only`	ReplicatedArm, RescueOnlyArm
`improved_arm_challenge_only`	ChallengeOnlyArm, ReplicatedArm
`improved_arm_challenge_late_rescue`	ChallengeRescueArm, ReplicatedArm
`improved_arm_challenge_early_rescue`	ChallengeRescueArm, PositiveControlArm, ReplicatedArm
`improved_cmp_validity`	CoveredComparison, InjuryValidityComparison, SingleFactorComparison
`improved_cmp_primary`	CoveredComparison, RescueEffectComparison, SingleFactorComparison
`improved_cmp_agent_alone`	AgentAloneComparison, CoveredComparison, SingleFactorComparison
`improved_cmp_positive_control`	CoveredComparison, PositiveControlComparison, RescueEffectComparison, SingleFactorComparison
`improved_cmp_window`	CoveredComparison, RescueTimingComparison, SingleFactorComparison

Closed-world checks:

- pass: every arm has one level of every factor (5 arms × 2 factors)
- pass: single factor: validity (challenge_only vs vehicle differ in challenge)
- pass: schedules matched: validity (administrations differ at 0 h, all belonging to the contrasted factor)
- pass: single factor: primary (challenge_late_rescue vs challenge_only differ in rescue_time)
- pass: schedules matched: primary (administrations differ at 48 h, all belonging to the contrasted factor)
- pass: single factor: agent_alone (rescue_only vs vehicle differ in rescue_time)
- pass: schedules matched: agent_alone (administrations differ at 48 h, all belonging to the contrasted factor)
- pass: single factor: positive_control (challenge_early_rescue vs challenge_only differ in rescue_time)
- pass: schedules matched: positive_control (administrations differ at 6 h, all belonging to the contrasted factor)
- pass: single factor: window (challenge_late_rescue vs challenge_early_rescue differ in rescue_time)
- pass: schedules matched: window (administrations differ at 6 h, 48 h, all belonging to the contrasted factor)
- information: full factorial (information, not required) (5 of 6 level combinations; missing: challenge=vehicle, rescue_time=h6)
- pass: interaction estimable (challenge vehicle/c085 × rescue_time none/h48)
- pass: every comparison has a decision rule (5 of 5)
- pass: positive control licenses the primary comparison (present)
- pass: replicates ≥ 3 per arm (vehicle 3, rescue_only 3, challenge_only 3, challenge_late_rescue 3, challenge_early_rescue 3)
- pass: units off the plate edge (15 interior wells)
- pass: all units on one plate (plate_P18)



The inferred hierarchy of the defined design concepts (solid) and the realization of the two designs' arms, comparisons and designs (dashed; blue: current, orange: improved).

Variants of the improved design

variant	most specific design concepts	closed-world checks failed
improved without the positive-control arm	AgentControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, ReplicatedDesign, RequiresImaging, SufficientlyInformativeDesign, WellContrastedDesign	positive control licenses the primary comparison
improved with one unit per arm	AgentControlledInjuryRescueDesign, DeviationAwareDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, PositiveControlledDesign, RequiresImaging, SufficientlyInformativeDesign, TherapeuticWindowDesign, WellContrastedDesign	replicates ≥ 3 per arm
improved without a pre-specified analysis	AgentControlledInjuryRescueDesign, FactorialDesign, HypothesisDiscriminatingDesign, InteractionAwareDesign, PositiveControlledDesign, ReplicatedDesign, RequiresImaging, StudyDesign, SufficientlyInformativeDesign, TherapeuticWindowDesign, WellContrastedDesign	every comparison has a decision rule

Inferred subsumptions not stated as axioms

- AdequatelyControlledInjuryRescueDesign ⊑ RequiresDosing
- AdequatelyControlledInjuryRescueExperiment ⊑ ValidatedInjuryRescueExperiment
- AgentControlledInjuryRescueDesign ⊑ RequiresDosing
- ComparativeDesign ⊑ RequiresCultivation
- ControlledDesign ⊑ RequiresCultivation
- ControlledExperiment ⊑ ComparativeExperiment
- DeviationAwareDesign ⊑ RequiresCultivation
- FactorialDesign ⊑ RequiresCultivation
- HypothesisDiscriminatingDesign ⊑ RequiresCultivation
- InjuryRescueDesign ⊑ RequiresDosing
- InjuryRescueExperiment ⊑ ControlledExperiment
- InteractionAwareDesign ⊑ RequiresCultivation
- NegativeControlArm ⊑ ArmSpecification
- PositiveControlledDesign ⊑ RequiresCultivation
- PreSpecifiedDesign ⊑ RequiresCultivation
- ReplicatedDesign ⊑ RequiresCultivation
- RequiresAbsorbance ⊑ RequiresEndpointHandling
- RequiresDosing ⊑ RequiresCultivation
- RequiresEndpointHandling ⊑ RequiresCultivation
- RequiresImaging ⊑ RequiresCultivation
- RequiresLuminescence ⊑ RequiresEndpointHandling
- RequiresMultipleEnvironments ⊑ RequiresCultivation
- SufficientlyInformativeDesign ⊑ RequiresCultivation
- TherapeuticWindowDesign ⊑ RequiresDosing
- TherapeuticWindowExperiment ⊑ InjuryRescueExperiment
- UntreatedArm ⊑ NegativeControlArm
- ValidatedInjuryRescueDesign ⊑ RequiresDosing
- ValidatedInjuryRescueExperiment ⊑ InjuryRescueExperiment
- VehicleControlArm ⊑ NegativeControlArm
- WellContrastedDesign ⊑ RequiresCultivation

Negative controls (each must be inconsistent)

- a well as part of an arm (parthood crosses kinds): inconsistent, as required
- a culture assigned to two arms (assignedTo is functional): inconsistent, as required
- an administration added to the untreated arm (closure): inconsistent, as required
- the graded hypothesis posits a loss of recovery capacity: inconsistent, as required
- the challenge agent also bears the rescue role (disjoint agents): inconsistent, as required
- one decision rule applied to two comparisons: inconsistent, as required
- an incubation function realized in an imaging assay: inconsistent, as required
- a factor level that is also an arm (disjoint plan parts): inconsistent, as required

17 ABox of the challenge-rescue experiment (excerpt)

StudyDesign(`challenge_rescue_design`)
Objective(`objective`)
GradedRecoveryHypothesis(`M1_graded`)
ThresholdRecoveryHypothesis(`M2_threshold`)
ProtectiveResponseHypothesis(`M3_protective`)
ChemicalAgent(`agent_challenge`)
ChallengeAgentRole(`role_challenge`)
ChemicalAgent(`agent_rescue`)
RescueAgentRole(`role_rescue`)
ToxicityDisposition(`toxicity`)
Factor(`factor_challenge`)
FactorLevel(`level_challenge_none`)
FactorLevel(`level_challenge_0p85au_0h`)
Factor(`factor_rescue`)
FactorLevel(`level_rescue_none`)
FactorLevel(`level_rescue_0p7au_48h`)
ArmSpecification(`arm_rescued`)
AdministrationSpecification(`admin_rescued_challenge_0h`)
AdministrationSpecification(`admin_rescued_rescue_48h`)
$(\leq 2\,\mathit{hasAdministrationSpec}.\mathsf{AdministrationSpecification})(\texttt{arm_rescued})$
OrganoidCulture(`unit_rescued_1`)
ExperimentalUnitRole(`unit_rescued_1_role`)
$(\leq 1\,\mathit{hasAssignedUnit}.\mathsf{ExperimentalUnit})(\texttt{arm_rescued})$
ArmSpecification(`arm_comparator`)
AdministrationSpecification(`admin_comparator_challenge_0h`)
$(\leq 1\,\mathit{hasAdministrationSpec}.\mathsf{AdministrationSpecification})(\texttt{arm_comparator})$
OrganoidCulture(`unit_comparator_1`)
ExperimentalUnitRole(`unit_comparator_1_role`)
$(\leq 1\,\mathit{hasAssignedUnit}.\mathsf{ExperimentalUnit})(\texttt{arm_comparator})$
ArmSpecification(`arm_control`)
$(\leq 0\,\mathit{hasAdministrationSpec}.\mathsf{AdministrationSpecification})(\texttt{arm_control})$
OrganoidCulture(`unit_control_1`)
ExperimentalUnitRole(`unit_control_1_role`)
$(\leq 1\,\mathit{hasAssignedUnit}.\mathsf{ExperimentalUnit})(\texttt{arm_control})$
$(\leq 3\,\mathit{hasArm}.\top)(\texttt{challenge_rescue_design})$
ComparisonSpecification(`cmp_validity`)
$(\leq 1\,\mathit{contrasts}.\top)(\texttt{cmp_validity})$
ComparisonSpecification(`cmp_primary`)
$(\leq 1\,\mathit{contrasts}.\top)(\texttt{cmp_primary})$
$(\leq 2\,\mathit{hasComparisonSpec}.\top)(\texttt{challenge_rescue_design})$
Viability(`viability`)
ReadoutSpecification(`readout`)
ImagingReadout(`readout`)
OrganoidSize(`organoid_size`)
AnalysisPlan(`analysis_plan`)
ThresholdRule(`rule_validity`)
BayesianUpdateRule(`rule_primary`)
Plate(`plate_P17`)
Well(`well_A01`)
Well(`well_A02`)
Well(`well_A05`)
Investigation(`experiment_challenge_rescue`)
CultivationPhase(`phase_cultivation`)
ObservationPhase(`phase_observation`)
AnalysisPhase(`phase_analysis`)
Administration(`step_x1`)
Administration(`step_x2`)
Incubation(`step_g1`)
Administration(`step_x3`)
Incubation(`step_g2`)
ImagingAssay(`step_o1`)
Image(`image_rescued`)
ImagingAssay(`step_o2`)
Image(`image_comparator`)
ImagingAssay(`step_o3`)
Image(`image_control`)
ImageAnalysis(`step_i1`)
SizeDatum(`size_rescued`)
ImageAnalysis(`step_i2`)
SizeDatum(`size_comparator`)
ImageAnalysis(`step_i3`)
SizeDatum(`size_control`)
StatisticalComparison(`step_v1`)
EffectEstimate(`effect_validity`)
StatisticalComparison(`step_a1`)
EffectEstimate(`effect_primary`)
ModelUpdate(`step_a1`)
PosteriorDistribution(`posterior`)
Prediction(`prediction_M1_graded`)
Prediction(`prediction_M2_threshold`)
Prediction(`prediction_M3_protective`)
Conclusion(`conclusion_answer`)
Conclusion(`conclusion_rescue_effect`)
Conclusion(`conclusion_challenge_effect`)
$\mathit{hasObjective}(\texttt{challenge_rescue_design}, \texttt{objective})$
$\mathit{isAbout}(\texttt{objective}, \texttt{M1_graded})$
$\mathit{isAbout}(\texttt{objective}, \texttt{M2_threshold})$
$\mathit{isAbout}(\texttt{objective}, \texttt{M3_protective})$
$\mathit{bearerOf}(\texttt{agent_challenge}, \texttt{role_challenge})$
$\mathit{bearerOf}(\texttt{agent_rescue}, \texttt{role_rescue})$

bearerOf(agent_challenge, toxicity)
hasFactor(challenge_rescue_design, factor_challenge)
hasLevel(factor_challenge, level_challenge_none)
hasLevel(factor_challenge, level_challenge_0p85au_0h)
hasFactor(challenge_rescue_design, factor_rescue)
hasLevel(factor_rescue, level_rescue_none)
hasLevel(factor_rescue, level_rescue_0p7au_48h)
hasArm(challenge_rescue_design, arm_rescued)
atLevel(arm_rescued, level_challenge_0p85au_0h)
atLevel(arm_rescued, level_rescue_0p7au_48h)
hasAdministrationSpec(arm_rescued, admin_rescued_challenge_0h)
usesAgent(admin_rescued_challenge_0h, agent_challenge)
hasAdministrationSpec(arm_rescued, admin_rescued_rescue_48h)
usesAgent(admin_rescued_rescue_48h, agent_rescue)
bearerOf(unit_rescued_1, unit_rescued_1_role)
assignedTo(unit_rescued_1, arm_rescued)
hasArm(challenge_rescue_design, arm_comparator)
atLevel(arm_comparator, level_challenge_0p85au_0h)
atLevel(arm_comparator, level_rescue_none)
hasAdministrationSpec(arm_comparator, admin_comparator_challenge_0h)
usesAgent(admin_comparator_challenge_0h, agent_challenge)
bearerOf(unit_comparator_1, unit_comparator_1_role)
assignedTo(unit_comparator_1, arm_comparator)
hasArm(challenge_rescue_design, arm_control)
atLevel(arm_control, level_challenge_none)
atLevel(arm_control, level_rescue_none)
bearerOf(unit_control_1, unit_control_1_role)
assignedTo(unit_control_1, arm_control)
hasComparisonSpec(challenge_rescue_design, cmp_validity)
compares(cmp_validity, arm_comparator)
compares(cmp_validity, arm_control)
contrasts(cmp_validity, factor_challenge)
hasComparisonSpec(challenge_rescue_design, cmp_primary)
compares(cmp_primary, arm_rescued)
compares(cmp_primary, arm_comparator)
contrasts(cmp_primary, factor_rescue)
licenses(cmp_validity, cmp_primary)
hasReadout(challenge_rescue_design, readout)
measures(readout, organoid_size)
isProxyFor(readout, viability)
hasAnalysisPlan(challenge_rescue_design, analysis_plan)
hasDecisionRule(analysis_plan, rule_validity)
appliesTo(rule_validity, cmp_validity)
hasDecisionRule(analysis_plan, rule_primary)
appliesTo(rule_primary, cmp_primary)
isAbout(rule_primary, M1_graded)
isAbout(rule_primary, M2_threshold)
isAbout(rule_primary, M3_protective)
hasWell(plate_P17, well_A01)
contains(well_A01, unit_rescued_1)
hasWell(plate_P17, well_A02)
contains(well_A02, unit_comparator_1)
hasWell(plate_P17, well_A05)
contains(well_A05, unit_control_1)
executes(experiment_challenge_rescue, challenge_rescue_design)
hasPhase(experiment_challenge_rescue, phase_cultivation)
hasPhase(experiment_challenge_rescue, phase_observation)
hasPhase(experiment_challenge_rescue, phase_analysis)
precedes(phase_cultivation, phase_observation)
precedes(phase_observation, phase_analysis)